

Evidence-Constrained Diagnostic Assistance for Weakly Labeled Neutron Measurement Data: A Tool-Using Workflow and Retrieval Robustness Study

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Abstract

Neutron-instrumentation records from nuclear power plants are high-volume and only weakly labeled, whereas their interpretation must remain traceable to authoritative technical evidence. We present an evidence-constrained diagnostic-assistance workflow rather than an autonomous fault classifier. A zero-prior, tool-using agent first converts neutron-current CSV records into an auditable condition report by applying deterministic probes for data health, morphology, anomalies, lag, extremes, and spatial snapshots. A compact structured alert summary then drives multi-stage retrieval over nuclear instrumentation and control manuals, enabling a citation-bearing diagnostic memo. We evaluate the retrieval layer under controlled expansion of a fixed IAEA core library with same-domain supplementary documents, using Chinese and English queries, chunking configurations, and both strict chunk and document-level relevance criteria. In the evaluated setup, cosine similarity can increase while the priority of pre-designated IAEA sources decreases; short Chinese queries are especially sensitive to corpus expansion, whereas the tested chunk-size and overlap changes have a smaller effect. The results show that similarity alone is insufficient to govern evidence retrieval in this setting and motivate terminology-rich queries, explicit source-priority policies, and human review of diagnostic conclusions.

Keywords: neutron instrumentation, online monitoring, diagnostic assistance, retrieval-augmented generation, evidence traceability, retrieval robustness

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1. Introduction

Nuclear power plants continuously acquire neutron and instrument-channel measurements for surveillance and condition monitoring. These records can reveal changes in detector behavior, signal integrity, and operating regimes, but are often sparsely labeled and difficult to align with a single sample-level fault taxonomy. Technical interpretation therefore continues to rely on authoritative guidance for online monitoring (OLM) and instrument-channel assessment (International Atomic Energy Agency, 2008a,b; Hashemian, 2011). Neutron-noise analysis further provides a non-intrusive route to studying reactor and instrumentation phenomena (Pázsit and Demazière, 2010; Demazière et al., 2022). Although data-driven diagnostic methods can be effective for predefined scenarios, their assumptions of fixed feature schemas and labeled fault classes are not always met by plant archives (Tambouratzis et al., 2020; Elbordany et al., 2024).

Large language models (LLMs) can help organize heterogeneous observations and technical knowledge, but free-form generation is unsuitable as a decision authority in a safety-relevant setting. Retrieval-augmented generation (RAG) conditions an answer on external evidence (Lewis et al., 2020); in nuclear applications, its value depends on whether the retrieved manuals genuinely support the statement being made. Recent nuclear RAG studies emphasize factuality, provenance, and failure modes such as incorrect retrieval and unsupported interpretation (Reeves et al., 2024; Lin et al., 2025; Mapes et al., 2025; Ndum et al., 2026). A practical open question is how to connect measurement-side findings to that evidence layer without asking an LLM to infer a root cause directly from a raw CSV file.

This work addresses that question through an evidence-constrained diagnostic-assistance workflow. A tool-using agent observes raw neutron-current series with minimal diagnostic priors and writes an auditable condition report. The report is then compressed into a structured alert summary that drives manual retrieval and citation-bearing explanation. The intended output is not an automatically confirmed fault diagnosis: it distinguishes observed data phenomena from manual-supported candidate interpretations, both of which remain subject to engineering review.

The reliability of this workflow depends critically on retrieval. A manual library may need to include both a small, pre-designated authoritative set

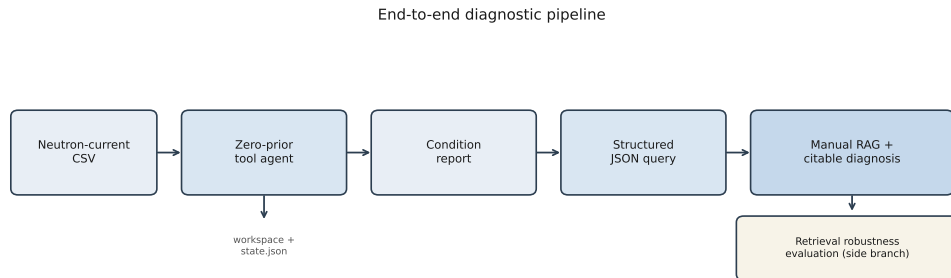


Figure 1: Evidence-constrained workflow. Deterministic probes turn a neutron-current CSV into an auditable condition report; a structured alert summary retrieves manual evidence for a citation-bearing diagnostic memo. Engineering review remains outside the automated chain.

and a larger set of same-domain supplements. Adding relevant supplements can improve apparent nearest-neighbor similarity while reducing the priority of the authoritative sources specified by a deployment policy. This tension is especially important for bilingual technical terminology and for overlap-based chunking, where a useful passage may not have the identical chunk identifier used as a query probe.

The contributions are threefold. First, we develop a zero-prior, tool-using workflow with a persistent workspace contract for converting neutron-current CSV data into auditable observations. Second, we introduce report-driven multi-stage RAG that maps structured observations to citable manual evidence. Third, we evaluate evidence retrieval under controlled core-supplement expansion, query language, chunking, and strict versus document-level relevance. The resulting evidence is used to characterize deployment boundaries rather than to claim autonomous fault identification. Figure 1 summarizes the workflow.

2. System design

2.1. Data description

Two complementary resources are used throughout this study: plant neutron-current measurements for agent-side diagnosis, and a curated nuclear I&C manual corpus for manual-grounded RAG and retrieval evaluation.

Neutron-current measurement archive.. The underlying measurement archive is organized into four groups of multi-channel neutron-current CSV records

(hereafter Groups A1, A2, B1, and B2), corresponding to different instrumentation banks in the same plant campaign. In total, the archive contains 42 series files, as summarized in table 1.

Table 1: Composition of the neutron-current measurement archive.

Group	Number of series	Channels per series	Columns per series
A1	11	7	8
A2	11	7	8
B1	10	7	8
B2	10	7	8
Total	42	—	—

Each series stores a time column together with seven neutron-current channels (eight columns overall); channel identifiers follow plant tagging conventions (e.g., RIC...MA_VER). Within each group, the seven detectors are arranged in a circular (circumferential) layout around the reactor, so neighboring channels correspond to adjacent angular positions rather than to an arbitrary listing order in the CSV. Representative series contain on the order of 6×10^5 samples at a nominal interval of 1 s and cover multi-day windows.

Importantly, timestamps in the *raw* CSV files are only approximately chronological: local disorders and non-monotonic segments appear, so the records are not strictly sorted in increasing time order as exported from the plant historian. The agent therefore begins with an explicit chronological sort before any downstream probe is applied.

Plant-side operational annotations and fault-related records exist for parts of this campaign; however, they are heterogeneous across sources and are not aligned into a uniform, sample-level fault taxonomy that could be tabulated without substantial manual curation. We therefore do *not* report fault-label counts in this paper. Instead, the zero-prior agent treats channel tags as opaque identifiers and does not consume fault labels during analysis; any available annotations are reserved for future supervised benchmarking rather than for the present pipeline.

For the system demonstration and case studies in section 4, we focus on a working subset drawn from Groups A1 and A2, processed independently in workspace-local analyses. The same agent protocol applies unchanged to the remaining series in the archive.

Nuclear I&C manual corpus. The knowledge corpus comprises 59 technical documents converted from PDF to Markdown for indexing and is partitioned into a *core* pool and a *non-core* (supplementary) pool. The core pool contains 13 IAEA manuals, including the IAEA OLM reports on instrument-channel monitoring and on process/component condition monitoring and diagnostics (International Atomic Energy Agency, 2008a,b), together with related IAEA guidance on I&C design, digital modernization, maintenance/surveillance, and operational limits and procedures. The non-core pool contains the remaining 46 same-domain manuals drawn from NRC, EPRI/DOE, OECD/NEA, and related laboratory or topical materials on sensors, noise, and core/instrument monitoring. These non-core documents are *not* treated merely as noise to be rejected; rather, they are topically related supplements that enlarge the searchable knowledge base in a controlled way. A central experimental question of this work is therefore the *core–non-core relationship*: how retrieval behavior changes as the ratio of core to non-core documents varies—equivalently, how large a non-core supplement can be admitted for a fixed core set before target (core) evidence is diluted. This controlled expansion underpins the corpus-scope and self-retrieval experiments in section 3.

2.2. Zero-prior diagnostic agent

2.2.1. Design objective

The measurement-side goal is to turn a raw neutron-current CSV into a structured, human-readable condition report without assuming a labeled fault library, a fixed channel-to-physics map, or handcrafted frequency bands. Channel tags are treated as opaque identifiers; statistical morphology, redundancy, anomalies, and spatiotemporal cues are discovered from the series themselves. This “zero-prior” stance matches the archive described in section 2.1, where fault annotations are heterogeneous and are deliberately not consumed at inference time.

2.2.2. Architecture and control loop

The agent couples a local LLM planner with a deterministic tool router (Yao et al., 2023). The planner proposes tool calls; a Python router executes registered analyzers and returns concise status messages that update a persistent workspace blackboard (`state.json`). Only the entry probe may receive a raw file path; thereafter every probe must take a workspace directory alone and resolve the active CSV from the blackboard. Figure 2 summarizes the control flow.

After initialization, the planner is not allowed to restate free-text file paths. LLM tool use remains brittle: models may invent nonexistent APIs, pass malformed arguments, or generate fluent but unfaithful content (Patil et al., 2024; Schick et al., 2023; Ji et al., 2023). The workspace contract therefore keeps authoritative paths and trusted state on the runtime side, while the planner selects only the next registered probe; the failure modes that motivated these constraints are detailed in section 2.2.5.

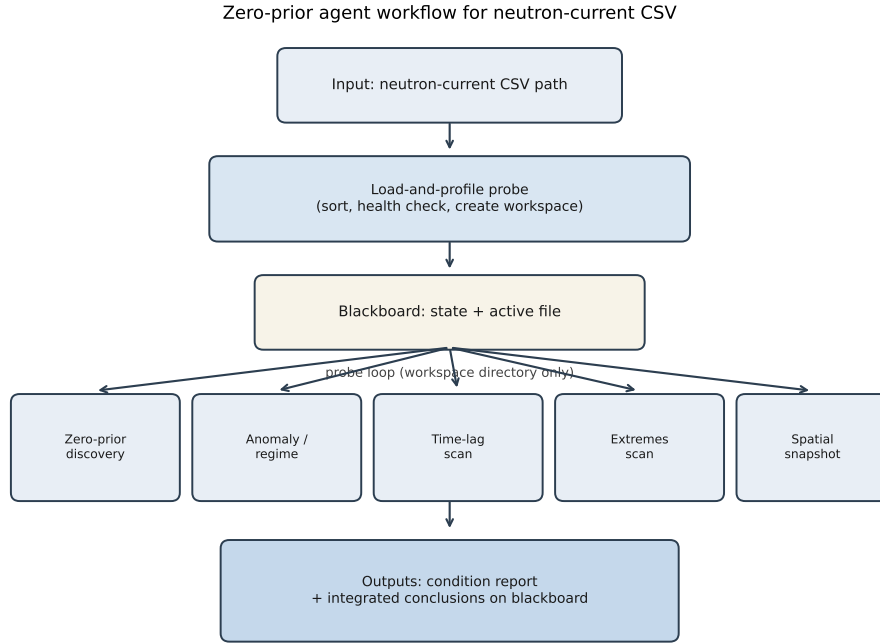


Figure 2: Zero-prior agent workflow for neutron-current CSV analysis. Profiling creates a workspace blackboard; later probes operate only on the workspace directory and accumulate a condition report.

2.2.3. Workspace initialization

Given a CSV path, the load-and-profile probe creates a dedicated workspace directory, parses the time column, sorts the series into increasing chronological order (addressing the raw non-monotonic exports noted in section 2.1), and writes a cleaned active file. It also records basic health indicators—shape, nominal sampling interval, missingness, and inactive channels—on the blackboard and appends an initialization block to the condition report.

Subsequent probes refuse to run if the blackboard is missing, which enforces a mandatory warm-up step.

2.2.4. Statistical discovery and anomaly probes

After initialization, the planner may invoke the probes listed in table 2. The zero-prior discovery probe performs a global longitudinal scan: uniqueness and coefficient of variation are used to assign coarse morphological roles (discrete/step-like, high-inertia, highly volatile, or ordinary continuous), while pairwise correlations flag near-redundant channel pairs ($|r| \geq 0.99$) and strongly co-moving groups. Given the circular detector layout, such collinearity often indicates angularly adjacent or functionally redundant sensors rather than coincidental numerical agreement.

The anomaly/regime probe then segments each channel into steady and transient regimes and enumerates spike episodes and isolated zero dropouts, summarizing which channels dominate each anomaly type. The time-lag scan computes cross-correlations across channels to separate lead-lag conduction links from essentially synchronous groups. The extremes scan localizes global maxima/minima and their timestamps; the spatial-snapshot probe freezes a multi-channel cross-section at a chosen time, which is useful after an extreme has been identified. Together, these probes accumulate evidence that can later be verbalized as spikes, dropouts, redundancy, and regime changes without supervised class labels.

Table 2: Diagnostic probes used by the neutron-current agent. The last column records the registered implementation identifier for reproducibility.

Probe	Description	Implementation ID
Load-and-profile	Creates workspace, sorts timestamps, records data-health indicators.	<code>load_and_profile_csv_tool</code>
Zero-prior discovery	Infers morphological roles and co-moving or near-redundant groups.	<code>zero_prior_discovery_tool</code>
Anomaly/region	Segments regimes; enumerates spikes and zero dropouts.	<code>anomaly_detector</code>
Time-lag scan	Estimates lead-lag versus synchronous groups.	<code>global_time_lag_scan_tool</code>
Extremes scan	Localizes global maxima/minima and timestamps.	<code>global_extremes_scan_tool</code>
Spatial snapshot	Multi-channel cross-section at a selected time.	<code>get_spatial_snapshot_tool</code>

2.2.5. Design process and adopted constraints

The architecture in section 2.2 is the outcome of an iterative, failure-driven design process rather than a single a priori specification. Early prototypes delegated almost all bookkeeping to the planner: each probe call restated an absolute CSV path, tool replies were expected to be nearly pure JSON, and the resulting condition text was passed almost unchanged into downstream diagnosis. Under a local mid-size LLM these assumptions failed repeatedly on multi-step neutron-current analyses. Table 3 summarizes the issues retained as hard constraints in the final protocol; the remainder of this subsection elaborates the failure mechanisms that motivate those rows.

Table 3: Design issues encountered in early agent prototypes and constraints adopted in the final measurement-side protocol.

Design issue	Adopted constraint
Path restatement errors across tool steps (wrong folder; raw vs. sorted copy)	Only load-and-profile accepts a CSV path; later probes take a workspace directory and resolve the active file from the blackboard.
Dropped or malformed tool JSON from local LLMs	Strip Markdown fences; extract the first brace-balanced JSON object/array; accept only registered probe identifiers.
Oversized condition text for downstream reasoning	Retain a human-readable report; drive RAG from a compact structured alert summary.
Ungrounded diagnostic prose without manuals	Build queries from the structured summary; require multi-stage, citation-bearing retrieval over the I&C corpus.

When every tool step re-emitted a free-text path, small transcription errors compounded over a probe chain: the planner confused campaign folders, mixed raw exports with chronologically sorted copies, or pointed later probes at a different series than the one just profiled. Because the runtime trusted the path argument, such mistakes executed silently and produced incoherent reports rather than explicit refusals. The adopted remedy is to treat the load-and-profile step as the sole path-accepting entry point. It creates a dedicated workspace, writes the cleaned active file, and records the authoritative location on the blackboard; thereafter probes accept only a workspace directory and resolve the series from that board (Yao et al., 2023; Patil et al., 2024). Path ownership thus moves from mutable model strings to a runtime contract, which also makes incomplete warm-up fail closed when the blackboard is missing.

A second class of failures arose at the action interface. Strict “JSON-only” prompting was brittle: local models frequently wrapped tool calls in Markdown fences, interleaved brief natural-language commentary, or truncated braces, so the router discarded otherwise valid actions; unrestricted free text made parsing equally unreliable. The present extractor therefore strips fences, searches for the first brace-balanced JSON object or array, and validates that the `action` field matches a registered probe identifier before execution (Yao et al., 2023; Schick et al., 2023). Invented analyzer names fail closed. This does not claim perfect structured generation; it treats imperfect local-model output as the default case and recovers the call when a well-formed object is present.

Even when the probe chain completed, an unfiltered multi-page condition narrative overloaded the manual-diagnosis stage: the LLM was asked to reason over lengthy prose while simultaneously selecting handbook passages, which encouraged vague, poorly grounded summaries. The design therefore separates audiences. The workspace retains a chronological, human-readable report for inspection and audit, while a compact structured summary—variable-wise alerts, spike/zero tallies, and collinearity or lag cues—is reserved as the machine interface to RAG (section 2.4). Retrieval queries are derived from that summary rather than from an operator’s free-text guess about which manual clause applies, and the synthesizer is required to attach numbered citations to retrieved chunks (Ji et al., 2023). Illustrative workspaces retained for this study show complete probe chains on representative A1/A2 series, while other series exhibit partial completion or redundant re-invocation of the same probe; these cases indicate that the workspace and JSON contracts are enforceable, yet planner discipline still affects coverage—a qualitative observation that motivates the protocol, not a supervised accuracy claim with fabricated recovery rates.

2.2.6. Calling protocol and outputs

Tool calls must appear as constrained JSON objects (`{"action": ..., "parameters": ...}`), optionally wrapped in Markdown fences; the runtime extracts the first balanced JSON object or array even when the LLM adds free-text commentary, mitigating the JSON-loss issue discussed above. Only registered probe identifiers (table 2) are accepted by the local router (Ollama `qwen2.5:7b` in our deployment), so invented analyzer names fail closed rather than executing an undefined path. Each successful probe appends narrative blocks to the workspace report and updates structured fields on the blackboard (e.g., morphological roles, spike/zero summaries, lag groups). The resulting condition report—together with optional JSON snapshots—forms the sole measurement-side interface to the RAG stage in section 2.4: downstream retrieval is driven by observed phenomena rather than by an operator’s free-text guess about which manual clause applies.

2.3. Manual corpus and index construction

Source PDFs are page-cropped when needed, converted to Markdown, chunked with a size/overlap grid, and embedded with `nomic-embed-text` into a LlamaIndex vector store (Liu, 2022). Indexing is applied to the corpus

described in section 2.1; evaluation varies which documents are admitted into the active retrieval pool while keeping the embedding model fixed.

2.4. Report-driven multi-stage RAG diagnosis

Condition reports are parsed into compact JSON (variable-wise alerts, spike summaries, topology/collinearity cues). Queries are constructed from this JSON rather than from short free-text alone. Retrieval proceeds in stages (phenomenon/mechanism passages, then redundancy/topology supplements) and the LLM composes a multi-part diagnosis with numbered citations (**[n]**, **Rn**) linked to retrieved chunks. Because diagnosis quality depends on retrieval, section 3 evaluates the knowledge layer separately.

2.5. Operational definitions and auditable interface

Let $\mathbf{X} = \{x_{t,c}\}$ denote a chronologically sorted multichannel record, where t indexes the sample and c indexes the detector channel. The system does not ask the planner to estimate these quantities. Deterministic probes compute them and write their results to the workspace blackboard; the LLM only selects registered probes and organizes the resulting evidence.

For each channel, the anomaly probe uses a centered rolling window W_t of 31 samples. Its local baseline and robust local deviation are

$$m_{t,c} = \text{median}_{\tau \in W_t} x_{\tau,c}, \quad d_{t,c} = \text{median}_{\tau \in W_t} |x_{\tau,c} - m_{\tau,c}|. \quad (1)$$

The implementation marks a sample as a spike candidate only when both a local and a global-amplitude condition hold:

$$\begin{aligned} I_{t,c}^{\text{spike}} &= \mathbb{I}(|x_{t,c} - m_{t,c}| > 5d_{t,c}) \mathbb{I}(|x_{t,c} - m_{t,c}| > 0.05R_c), \\ R_c &= \max_t x_{t,c} - \min_t x_{t,c}. \end{aligned} \quad (2)$$

Contiguous positive samples are merged into one spike episode. An isolated zero event is separately recorded by $I_{t,c}^{\text{zero}} = \mathbb{I}(x_{t,c} = 0)$. These rules define detected signal events, not physical fault labels.

For a channel pair (i, j) , the discovery probe computes Pearson correlation

$$r_{ij} = \frac{\sum_t (x_{t,i} - \bar{x}_i)(x_{t,j} - \bar{x}_j)}{\sqrt{\sum_t (x_{t,i} - \bar{x}_i)^2} \sqrt{\sum_t (x_{t,j} - \bar{x}_j)^2}}. \quad (3)$$

The current implementation flags near-redundancy at $r_{ij} \geq 0.99$ and strong co-movement for $0.85 \leq r_{ij} < 0.99$. Together with lag, extremes, and spatial-snapshot outputs, these quantities form a structured alert interface

$$\mathbf{a} = g(\mathbf{X}) = \{\mathbf{a}_{\text{health}}, \mathbf{a}_{\text{morph}}, \mathbf{a}_{\text{anom}}, \mathbf{a}_{\text{lag}}, \mathbf{a}_{\text{spatial}}\}, \quad (4)$$

which is passed to retrieval. This separation prevents free-form text from becoming the sole interface between measurement analysis and manual-grounded explanation.

3. Evaluation protocol

3.1. Evaluation objective

The evaluation assesses the reliability of the *evidence-retrieval layer*, not diagnostic accuracy or root-cause identification. We ask whether retrieval continues to honor a deployment policy that prioritizes a fixed IAEA core library when same-domain supplementary manuals are admitted, and how this behavior depends on query language, query formulation, and chunking.

3.2. Corpus and index configurations

The corpus contains a fixed core pool of 13 IAEA manuals and up to 46 same-domain supplementary manuals, for 59 documents in total. The supplementary pool is relevant nuclear instrumentation material; it is not treated as meaningless noise. Seventeen Markdown index configurations vary chunk size $\{800, 1000, 1200, 1500\}$ and overlap $\{50, 80, 100, 120, 150\}$ for valid pairs. The embedding model is fixed as `nomic-embed-text`.

3.3. Corpus-scope experiment

Let $\mathcal{D}_{\text{core}}$ be the fixed IAEA pool and $\mathcal{D}_{\text{sup}}^{(m)}$ the m admitted supplementary documents. Each retrieval pool is $\mathcal{D}_m = \mathcal{D}_{\text{core}} \cup \mathcal{D}_{\text{sup}}^{(m)}$. Four fixed Chinese and English nuclear terms or phrases are issued while m increases in increments. We report top- k cosine scores, IAEA priority@10, and the rank of the first IAEA hit. For a query q and its ordered retrieval list $R_k(q) = \{d_1, \dots, d_k\}$,

$$\text{Priority}@k(q) = \frac{1}{k} \sum_{i=1}^k \mathbb{I}(d_i \in \mathcal{D}_{\text{core}}). \quad (5)$$

This is the fraction of top- k chunks from the pre-designated IAEA core. It measures adherence to a *source-priority policy*; it is not a substitute for human relevance judgment, because a supplementary document may also be substantively relevant.

3.4. Self-retrieval experiment

IAEA chunks are sampled as probes. Queries are constructed from a first sentence, title, random window, or terminology-rich domain sentence. For a probe chunk n_j^* and retrieved set $R_k(q_j)$, strict and document-level recall are

$$\begin{aligned} \text{Recall@ } k_{\text{strict}} &= \frac{1}{N} \sum_{j=1}^N \mathbb{I}(n_j^* \in R_k(q_j)) , \\ \text{Recall@ } k_{\text{doc}} &= \frac{1}{N} \sum_{j=1}^N \mathbb{I}(\exists n \in R_k(q_j) : \text{doc}(n) = \text{doc}(n_j^*)) . \end{aligned} \tag{6}$$

We additionally report mean reciprocal rank. The distinction prevents overlap-induced near-duplicate chunks from being mischaracterized as entirely irrelevant.

3.5. Interpretation boundary

The experiments establish retrieval behavior for the corpus, embedding model, query set, and index grid studied here. They do not establish a universal ranking of languages or chunking strategies, nor do they evaluate the factual correctness of generated memos. Those questions require additional multilingual models, relevance annotation, and generation-side review.

4. Results and discussion

4.1. Case study: from observation to evidence

On representative A1/A2 workspaces, the measurement-side probes produce reports containing synchronized zero events, spike episodes, and strongly co-moving channel groups. For example, the A1_1 report identifies high-sensitivity morphology across channels, several highly collinear detectors, repeated spike episodes on one channel, and zero dropouts on others. Figure 3 displays downsampled A1_1 traces; all probe outputs are computed on the full-resolution series.

This result demonstrates that the workflow can produce an auditable observation record and use it to organize manual retrieval. It should not be interpreted as independent confirmation of a physical fault or a root cause. In the present study, the memo is an evidence-linked aid for subsequent engineering review; no uniformly aligned fault-label set is available for estimating end-to-end diagnostic accuracy.

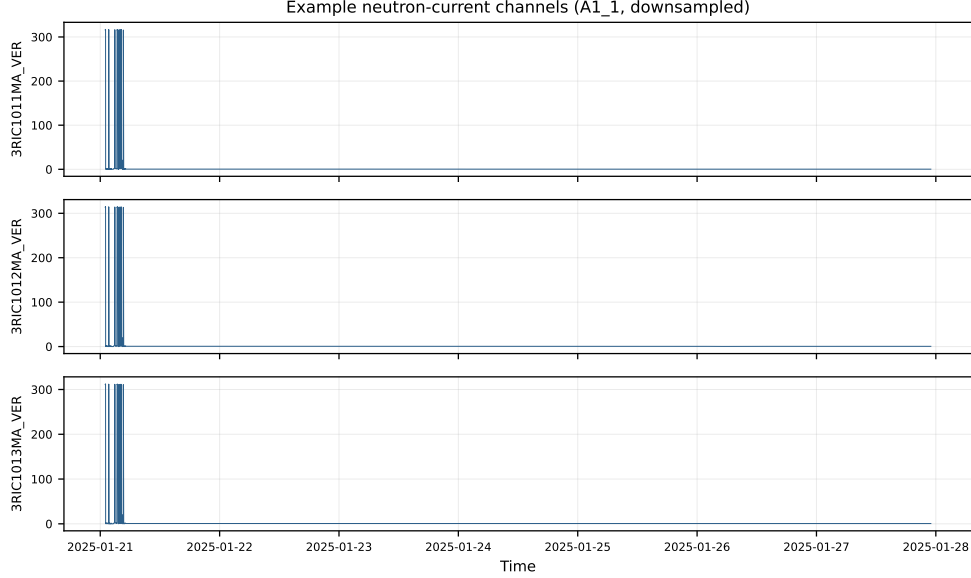


Figure 3: Example neutron-current traces from dataset A1_1 (three channels; downsampled for readability). The full-resolution series is analyzed by the workspace probes.

Figure 4 and Table 4 provide a second, tabulated case view from the checked-in A2_1 report snapshot. The seven channels contain 30–34 detected spike episodes (mean 32.4) and each contains 12 isolated zero events. The repeated zero timestamps across channels are a concrete example of why the report preserves cross-channel timing rather than converting events immediately into a component-fault label: the observation supports a synchronized signal-quality event, while its physical mechanism requires external review.

4.2. Similarity and source priority can diverge

The corpus-scope results show a consistent deployment risk: increasing the supplementary pool can raise cosine scores while reducing IAEA priority@10. Figure 5 and table 5 aggregate the phrase-query results over all 17 available index configurations. For Chinese, mean IAEA priority@10 falls from 1.00 in the IAEA-only pool to 0.10 ± 0.09 after five supplementary documents and to 0.00 after ten, while the mean top-1 cosine rises from 0.55 to 0.63. English is more gradual: priority@10 remains 0.86 ± 0.05 after five documents and is 0.38 ± 0.05 at 45 documents, while the corresponding cosine

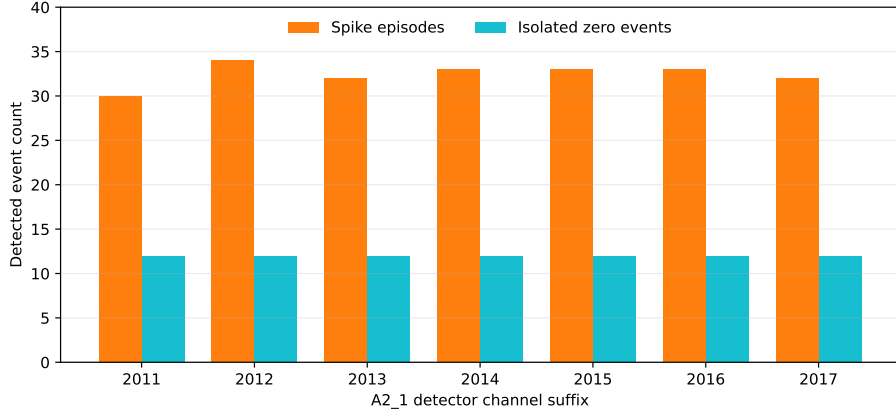


Figure 4: Detected event counts from the A2_1 condition-report snapshot. These are deterministic signal-event counts under eq. (2), not confirmed equipment-fault counts.

stays near 0.69–0.70.

These results do not imply that supplementary documents are irrelevant. Rather, when an application requires IAEA sources to be surfaced first, nearest-neighbor similarity alone is not enough to enforce that policy. Source metadata, filtering, or reranking should be considered before a generated memo is presented as authority-backed evidence.

4.3. Language and query formulation dominate the tested chunking range

Across the tested index grid, the qualitative language effect is larger than the variation induced by modest changes in chunk length and overlap. Short Chinese prompts are particularly vulnerable to source-priority collapse, whereas terminology-rich English queries retain more core material. This is a finding for the fixed embedding model and corpus, not a claim that one language is intrinsically less retrievable. It supports deriving detailed, terminology-rich queries from the condition report instead of issuing only an ultra-short operator phrase.

4.4. Strict and document-level relevance answer different questions

Self-retrieval exposes a systematic strict–document-level gap. Figure 6 aggregates first-sentence corpus-decay results over the available result directories and shows the same separation for both Recall@10 and MRR. In one 800/50 setting, IAEA-only Recall@10 is 0.7625 under exact-node matching

Table 4: A2_1 event summary generated from the checked-in condition-report snapshot. Counts describe detected signal events, not confirmed equipment faults.

Channel suffix	Spike episodes	Isolated zero events
2011	30	12
2012	34	12
2013	32	12
2014	33	12
2015	33	12
2016	33	12
2017	32	12
Mean	32.4	12.0

Table 5: Corpus-scope results for phrase queries, averaged across the 17 available index configurations. IAEA priority@10 is a source-priority measure, not a general relevance judgment.

Query language	Supplementary documents	IAEA priority@10	Mean top-1 cosine
English	0	1.00 ± 0.00	0.69 ± 0.00
English	5	0.86 ± 0.05	0.69 ± 0.00
English	10	0.90 ± 0.02	0.69 ± 0.00
English	45	0.38 ± 0.05	0.70 ± 0.01
Chinese	0	1.00 ± 0.00	0.55 ± 0.00
Chinese	5	0.10 ± 0.09	0.62 ± 0.01
Chinese	10	0.00 ± 0.00	0.63 ± 0.01
Chinese	45	0.00 ± 0.00	0.67 ± 0.00

and 0.8625 under document-level matching. The gap is expected when overlapping chunks preserve relevant content in a neighboring node. Conversely, terminology-rich domain sentences have substantially stronger document-level Recall@10 in the available results. Evaluation should therefore report both criteria: strict matching measures chunk re-identification, while document-level matching better reflects whether the correct manual was recovered.

4.5. Real CPU embedding-model comparison

To test whether the observed behavior is tied only to the historical embedding index, we ran a separate CPU benchmark over the checked-in 64-document corpus. This benchmark uses 240-word chunks with 24-word over-

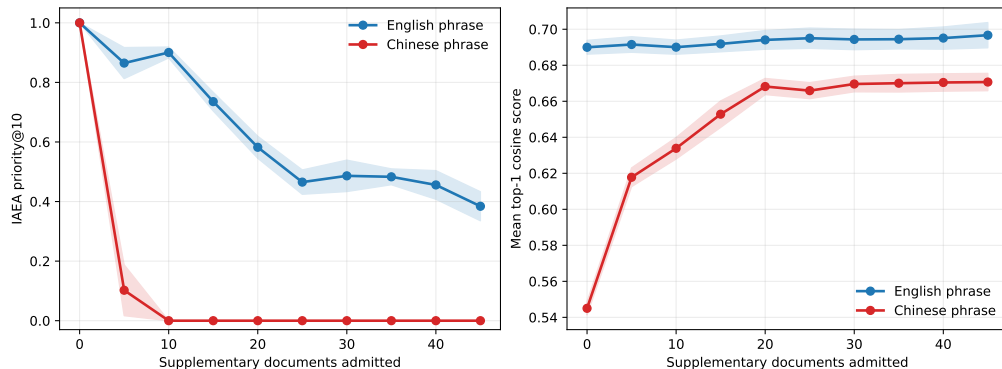


Figure 5: Phrase-query corpus-scope results averaged across 17 index configurations; shaded bands show one standard deviation across configurations. Source priority can decline while the nearest-neighbor cosine score rises.

lap (11,199 chunks), the same 13-document IAEA core and 51-document supplementary pool, the same four fixed bilingual queries, top-10 retrieval, and ten random expansion trials. It compares a multilingual MiniLM encoder with an English-oriented MiniLM encoder; it is reported separately from the historical `nomic-embed-text` experiments because the chunk construction differs.

Figure 7 and table 6 show that embedding choice changes cosine scales and source-priority trajectories, but does not collapse the problem into one scalar. The multilingual model gives materially higher Chinese phrase cosine scores (0.62–0.64 versus 0.27 for the English-oriented model), while both models undergo source-priority decay as the supplementary pool grows. Thus multilingual semantic alignment and authoritative-source preference are complementary deployment requirements rather than interchangeable objectives.

4.6. Implications and limitations

The pipeline provides traceable observations and evidence retrieval, but it does not replace engineering judgment. The current evidence is limited by case-level rather than gold-standard diagnostic evaluation, one embedding model, a small fixed query set for the scope study, and no human assessment of generated citation support. A next validation stage should separately assess observation correctness, passage support for each memo claim, and the usefulness of the resulting memo to domain experts.

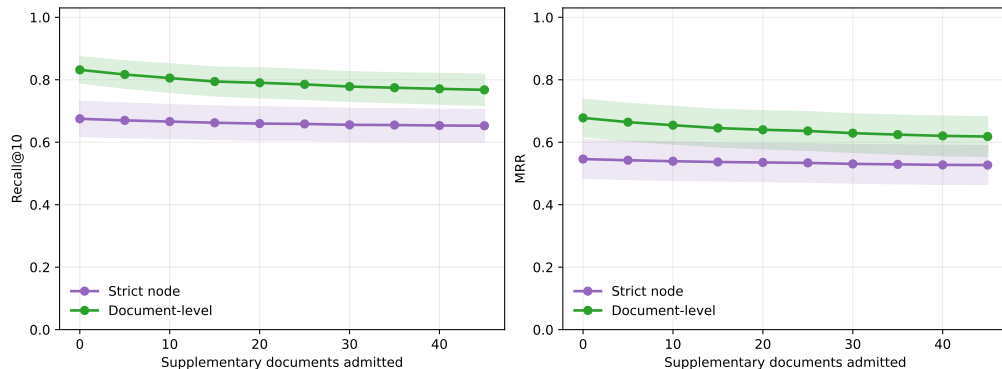


Figure 6: First-sentence self-retrieval under corpus expansion, aggregated over available result directories. Shaded bands show one standard deviation across configurations. Document-level relevance remains higher than exact-node matching because overlapping chunks can preserve relevant text within the same document.

5. Conclusions

We presented an evidence-constrained workflow for diagnostic assistance on weakly labeled neutron-current data. The workflow uses deterministic analysis probes under a persistent workspace contract to produce auditable observations, then turns a structured alert summary into manual-grounded, citation-bearing explanatory support. Its output is deliberately framed as an aid to engineering review rather than an autonomous fault decision.

The retrieval study identifies a practical governance issue: for the evaluated corpus and embedding model, cosine similarity can improve while the priority of pre-designated IAEA sources declines as same-domain supplementary manuals are added. Short Chinese queries are particularly sensitive in the tested setting, and document-level relevance is more forgiving than exact chunk identity under overlapping segmentation. These results favor terminology-rich report-derived queries and explicit source policies over similarity-only deployment choices.

Future work will validate the workflow with expert review of observed phenomena and citation support, construct a small relevance-annotated query set, and assess multilingual encoders, rerankers, metadata controls, and generation-side factuality. Such evaluation is required before any operational use beyond supervised diagnostic assistance.

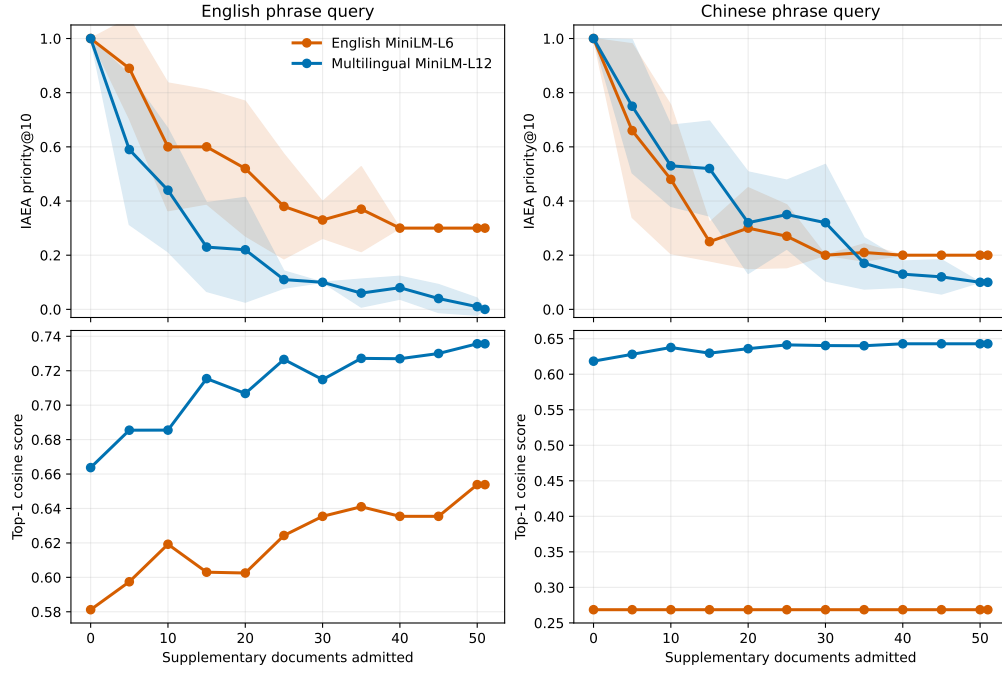


Figure 7: Actual CPU embedding benchmark on the same 64-document corpus. Curves compare a multilingual MiniLM encoder and an English-oriented MiniLM encoder; bands show standard deviation over random supplementary-document selections. This is a newly run benchmark, not an extrapolation of the historical index.

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Table 6: Real embedding-model comparison under the new CPU benchmark (64 documents, 13 IAEA core documents, 11,199 chunks, 10 random trials after corpus expansion). This benchmark uses 240-word chunks and is reported separately from the historical nomic-embed-text index.

Model	Query language	Supplementary documents	IAEA priority@10	Top-1 cosine
Multilingual MiniLM-L12	English	0	1.00 ± 0.00	0.66
Multilingual MiniLM-L12	English	10	0.44 ± 0.23	0.69
Multilingual MiniLM-L12	English	51	0.00 ± 0.00	0.74
Multilingual MiniLM-L12	Chinese	0	1.00 ± 0.00	0.62
Multilingual MiniLM-L12	Chinese	10	0.53 ± 0.15	0.64
Multilingual MiniLM-L12	Chinese	51	0.10 ± 0.00	0.64
English MiniLM-L6	English	0	1.00 ± 0.00	0.58
English MiniLM-L6	English	10	0.60 ± 0.24	0.62
English MiniLM-L6	English	51	0.30 ± 0.00	0.65
English MiniLM-L6	Chinese	0	1.00 ± 0.00	0.27
English MiniLM-L6	Chinese	10	0.48 ± 0.27	0.27
English MiniLM-L6	Chinese	51	0.20 ± 0.00	0.27

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